Robust Design & Product Specification

Specification of Products and Services · 7 sessions · 4th semester Industrial Design

Product Specification Report

Production Manual

Product Landing Page

What the course is about

Students transform previous project work into technical product documentation and a public-facing product page.

1 **Technical depth**

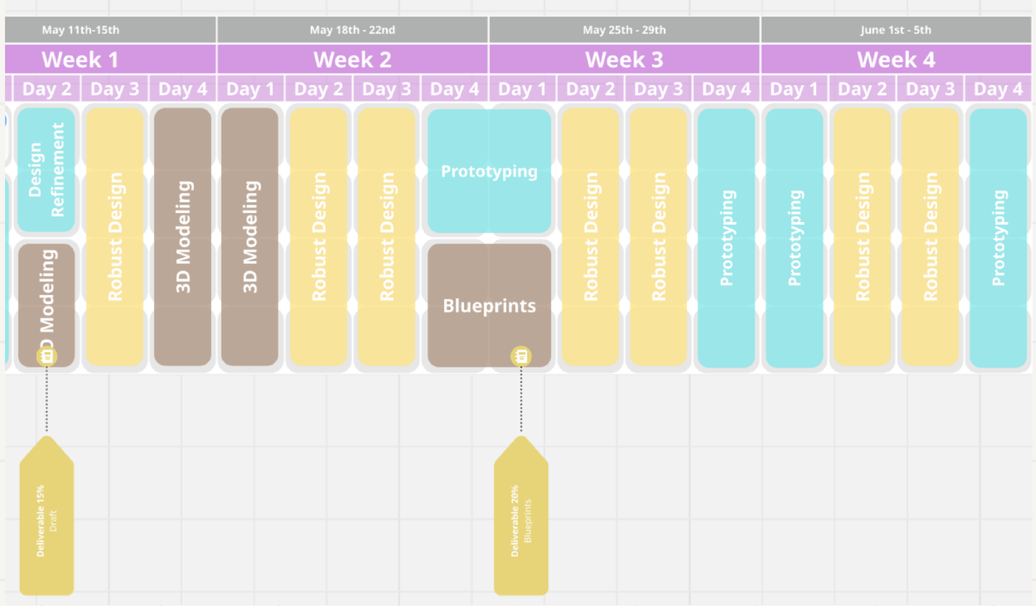
Build a complete Product Specification Report / Production Manual with requirements, materials, drawings, BOM, ergonomics, production and sustainability.

2 **Communication clarity**

Translate the most important technical content into a concise, attractive landing page that promotes the product.

3 **Progressive integration**

Each session generates one section of the final report, so the final delivery is assembled step by step.



Previous process map and schedule become inputs, not duplicated content.

The course closes the loop: concept → specification → production readiness → public communication.

Final deliverables

Two connected outputs: one deep technical document and one concise promotional website.

A

Extended Product Specification Report

Complete technical document: concept, requirements, robust design, materials, CMF, technical/productive specs, drawings, BOM, ergonomics, MVP, sustainability, normativity and references.

B

Product Landing Page

A visual one-page website that promotes the product: problem, solution, key benefits, how it works, features, technical highlights, sustainability, gallery and report download link.

Report = technical evidence

Landing page = attractive communication

The website does not replace the report. It synthesizes it and makes the product easy to understand, evaluate and promote.

Course workflow

NotebookLM and AI tools help students compile, organize and convert project evidence into final outputs.

1 → 2 → 3 → 4 → 5

1 Collect

Previous work, images, feedback, references

2 Organize

NotebookLM + project folder

3 Draft

Report sections + landing copy

4 Publish

Gamma / Canva / Notion / Google Sites

5 Present

Critique and final delivery

Each session closes with a concrete piece of the final report and a matching website section.

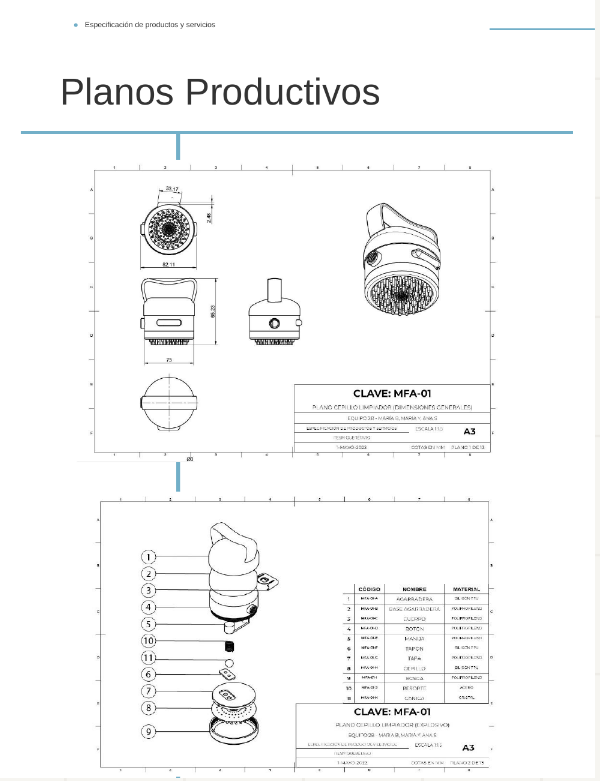
Seven-session roadmap

The course is structured as a progressive build-up of the final report and landing page.

1	**Diagnosis + setup**	Report index + landing wireframe
2	**Concept + requirements**	Overview, value proposition, requirements
3	**Materials + CMF**	Material sheets + technical specifications
4	**Production + robust design**	Production flow + risk matrix
5	**Drawings + BOM**	Drawing package + cost/supplier table
6	**Ergonomics + sustainability**	Use diagrams + MVP + life cycle
7	**Final integration**	Report + published landing page

Reference quality from previous courses

Past deliverables show that the final report should be a production manual, not only a visual portfolio.



Drawings


Catalog - Sistema de captación pluvial

Date - 25 Apr 2021; 01:51:22 GMT
Updated by - a01351633@itesm.mx
Created by - a01351633@itesm.mx

ne	Description	Material	Quantity On Hand	Price	Supp
nque	Conector de tanque de agua de latón sólido	Brass	3	$29.76	Yuhuan Aqua-Mige Co., Ltd.
ctor de tanque	Empaque conector de tanque de agua	Rubber	6	No aplica	Yuhuan Aqua-Mige Co., Ltd.
acenamiento	Módulo de almacenamiento para captación pluvial	Polyethylene, High Density	1	$40.09	Gansu Longchang Group Co., LTD
ueo de ique	Tuerca de bloqueo de tanque de agua	Brass	3	No aplica	Yuhuan Aqua-Mige Co., Ltd.
	Tapa superior para módulo de almacenamiento	Polyethylene, High Density	1	$12.85	Gansu Longchang Group Co., LTD
	Tapa inferior para módulo de almacenamiento	Polyethylene, High Density	1	$20.11	Gansu Longchang Group Co., LTD
lo de o	Empaque para módulo de almacenamiento para captación pluvial	Rubber	2	$0.44	Jinjiang Gangshen Export Co., LTD
	LLave de nariz 3/4	Brass	1	$30.95	Fu San Valve (dong
llave de nariz	Empaque para llave de nariz del módulo de almacenamiento	Rubber	2	$0.6	Xiamen Lindas Har Co., LTD
ve de nariz	Tuerca para llave de nariz 3/4	PVC-Piping	1	$16.66	McMASTER-CARR
	Tapón roscado 3/4	Brass	3	$14.28	Taizhou Kingston V
	Broche hebilla para accesorio jardín vertical	Nylon	8	$11.12	Hangzhou Minshi C Accessories Co., Ltc
vertical	Accesorio jardín vertical para sistema de captación pluvial	Felt	1	$39.27	Changshu Jiaming Co., Ltd.

BOM + costs

Life cycle

The new course adds a product landing page to communicate this technical work attractively.

Session 1: Project Diagnosis and Final Deliverables Setup

Start with what students already have and map what is missing for a professional final package.

Objective

Define the structure of the final report and landing page; organize existing project evidence; identify gaps and risks.

Report output

Project overview, initial report index, technical gap analysis and robust design diagnosis.

Website output

Landing page wireframe with hero, problem, solution, benefits and key visuals to collect.

Session 1 guiding question

What evidence do we already have, and what do we still need to prove the product can be produced, used, validated and promoted?

By the end of the class, every team should know exactly what to build next.

Session 1 agenda

Four hours focused on setup, diagnosis and first draft of the final structure.

00:00–00:25	**Course intro**	Deliverables, workflow, examples and evaluation logic
00:25–01:05	**Reference review**	Analyze previous manuals and extract quality criteria
01:05–01:55	**Project inventory**	Collect sources, images, models, feedback and existing evidence
02:10–03:05	**AI-assisted diagnosis**	Use NotebookLM to detect missing information and weak sections
03:05–03:45	**Report + landing setup**	Create report index and landing page wireframe
03:45–04:00	**Exit review**	Submit Session 1 evidence and next-step checklist

Activity 1: Project evidence inventory

Students organize the materials that will feed the report and landing page.

1 **Concept**

brief, problem, user, sketches, moodboard

2 **Product**

renders, 3D screenshots, prototype photos

3 **Technical**

materials, dimensions, drawings, parts, processes

4 **Validation**

feedback, tests, ergonomic notes, MVP limits

5 **Business**

market requirements, price target, suppliers, costs

6 **Impact**

sustainability, life cycle, regulations, references

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Activity 2: AI-assisted gap analysis

NotebookLM is used to compare project evidence against the expected final report.

Prompt for NotebookLM

Review all sources in this notebook and identify what information is missing to complete a professional Product Specification Report and Production Manual. Organize the missing information by: concept, requirements, materials, technical specifications, productive specifications, drawings, BOM, ergonomics, MVP, sustainability, normativity and references.

Output 1
Missing information map

A list of gaps organized by report section.

Output 2
Risk and robust design matrix

Possible failures, causes, consequences and design improvements.

Output 3
Next-step checklist

What each team must collect, draw, test or justify before Session 2.

Session 1 deliverable / evidence

What students submit at the end of the first class.

✓ Project folder

All available project files organized and named

✓ NotebookLM notebook

Sources uploaded and ready for AI-assisted review

✓ Report index

Draft structure of the extended technical report

✓ Landing wireframe

Hero, problem, solution, benefits and visual plan

✓ Gap analysis

Missing information and technical risks identified

✓ Next steps

Specific tasks assigned to team members

Next class starts from the requirements table and value proposition.